

Answer Key — Biology Grade 11

Sub-Strand 1.3: Taxonomy II • Lesson 2: Kingdom Plantae — Bryophyta

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
C	A	C	C	D	D	A

USE AFTER: PREDICT PHASE — Predict: private prediction sheet and Think-Pair-Share on 'Is moss more like maize, grass, or fern?'

Q1. Before examining moss, a learner predicts: "Moss is like maize. It has true roots and flowers." Which statement about moss is correct?

- A. Moss is a fungus, like a mushroom, so it has no green parts
- B. Moss has true roots, but they are shorter than those of maize
- C. Moss has no flowers and reproduces using spores, not seeds**
- D. Moss has flowers but they are too small to see

Correct answer: C. Mosses (Division Bryophyta) have no flowers, no seeds and no true roots. They reproduce by spores formed in capsules. They are green plants, not fungi.

USE AFTER: OBSERVE PHASE — Observe: hand-lens examination of the moss base and comparison with the maize or bean root

Q2. A learner looks at the base of a moss plant with a hand lens and sees many thin, thread-like structures going into the soil. What are they and what is their main job?

- A. Rhizoids that anchor the plant**
- B. True roots that carry water up through xylem
- C. Root hairs that carry food down through phloem
- D. Spore capsules that release spores into the soil

Correct answer: A. Moss has rhizoids, not true roots. They mainly anchor the plant and do not transport water like the true roots of maize.

USE AFTER: OBSERVE PHASE — Observe: labelled drawing of one moss plant with title, magnification and label lines

Q3. A group makes a labelled biological drawing of one moss plant. Which drawing is done correctly?

- A. It is shaded to show colour, with label lines that cross each other so all labels fit
- B. It has curved label lines drawn over the structures to show exactly where each part is
- C. It has a title and magnification, uses a sharp pencil with single clear lines, and has straight label lines that do not cross**
- D. It has no title, so the drawing looks neat and uncluttered

Correct answer: C. A good biological drawing has a title and magnification, clear single lines with no shading, and straight label lines that do not cross.

USE AFTER: OBSERVE PHASE — Observe: table with columns Feature | Moss | Maize/Bean | Fern

Q4. A group compares moss with a maize seedling in a table. Which row of differences is correct?

- A. Moss: large size and true leaves. Maize: very small size and no leaves.
- B. Moss: flowers and seeds. Maize: spore capsules on stalks.
- C. Moss: rhizoids, no vascular tissue, spore capsules. Maize: true roots, xylem and phloem, seeds.**
- D. Moss: true roots and xylem. Maize: rhizoids and no vascular tissue.

Correct answer: C. Moss has rhizoids, lacks xylem and phloem, and makes spores in capsules. Maize has true roots, vascular tissue and seeds. The other rows swap or invent features.

USE AFTER: EXPLAIN PHASE — Explain: Think-Pair-Share 'Why does moss stay small and grow only in damp places?'

Q5. Maize carries water up its stem through xylem. Moss has no xylem or phloem. What does this lack of vascular tissue tell us about moss?

- A. It grows tall in dry places because it does not need water
- B. It cannot make food because it has no phloem
- C. It stays small because it has too many rhizoids
- D. It stays small because it cannot transport water and food over long distances**

Correct answer: D. Without xylem and phloem, water and food cannot be moved efficiently over long distances. Moss therefore stays small and lives where water is easily available. It still makes its own food by photosynthesis.

USE AFTER: EXPLAIN PHASE — Explain: 'How does it reproduce without seeds or flowers?' and the link to damp places near Kericho and Nairobi

Q6. A learner finds moss growing on a damp, shaded wall in Nairobi, but none on a dry, sunny rock nearby. Which statement best explains this and how the moss reproduces?

- A. Moss needs deep true roots to reach water, so it grows only on walls
- B. Moss needs bright sunshine to form seeds, so it avoids the dry rock
- C. Moss reproduces by flowers that open only in shade
- D. Moss needs water for fertilisation and has no vascular tissue to bring water from far away, so it lives in damp places and reproduces by spores**

Correct answer: D. Bryophytes need water for fertilisation and cannot transport water far without vascular tissue. They make spores in capsules on stalks, not seeds or flowers.

USE AFTER: MODEL BUILDING PHASE — Model: revising the Lesson 1 classification chart with a 'Bryophyta' category and criteria

Q7. A group is adding a 'Bryophyta' category to its classification model. Which criterion is most useful for deciding whether a plant belongs in the group?

- A. It has no true roots, stems or leaves, has rhizoids, and bears spore capsules on stalks**
- B. It is green in colour
- C. It is not used as food by people
- D. It is small enough to fit in your hand

Correct answer: A. Groups should be defined by shared, observable structural features. Colour, size alone and usefulness do not separate moss from other plants, since maize and fern are also green.